Measure Theoretic Probability
Lesson 20
Examples and Criteria for Almost Sure Convergence

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Outline

1. Two Examples of Almost Sure Convergence
2. New Random Variables from Sequences
3. Events Defined by Random Variables
4. Criteria for Almost Sure Convergence
5. Summary and Exercises

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Example 1: $X_n(\omega) = \omega^n$

Setup

$\Omega = [0,1]$, $\mathcal{F} = \mathcal{B}([0,1])$, $\mathbb{P}$ = Lebesgue measure (a probability measure, since the total length is 1). Let $X_n(\omega) = \omega^n$.

Candidate limit

$$X(\omega) = \begin{cases} 0, & 0 \le \omega < 1, \\ -3, & \omega = 1. \end{cases}$$

Convergence

- $0 \le \omega < 1$: $\omega^n \to 0 = X(\omega)$.
- $\omega = 1$: $X_n(1) = 1$ for all n, so $X_n(1) \to 1 \neq -3 = X(1)$.

So $A = \{\lim_n X_n = X\} = [0,1)$ and $\mathbb{P}(A) = 1$. Hence $X_n \xrightarrow{\text{a.s.}} X$.

Remark

The value -3 is arbitrary: any value at $\omega = 1$ works, because $\{1\}$ has Lebesgue measure 0.

Is X_n Really a Random Variable?

Check via rays

Fix $x \in \mathbb{R}$ and compute $\{X_n \le x\} = \{\omega \in [0,1] : \ \omega^n \le x\}$. (Here ω always ranges over Ω.)

- $x < 0$: no ω qualifies, so $\{X_n \le x\} = \varnothing$.
- $0 \le x < 1$: $\omega^n \le x \iff \omega \le x^{1/n}$, so $\{X_n \le x\} = [0, x^{1/n}]$.
- $x \ge 1$: every $\omega \in [0,1]$ qualifies, so $\{X_n \le x\} = [0,1]$.

Conclusion

In all three cases the set is a Borel subset of $[0,1]$, so it lies in $\mathcal{F}$. By the rays criterion, each X_n is a random variable.

Example 2: $X_n(\omega) = \omega^n + \omega$

Setup

Same space $([0,1], \mathcal{B}([0,1]), \text{Lebesgue})$. Let $X_n(\omega) = \omega^n + \omega$ and $X(\omega) = \omega$.

Convergence

- $0 \leq \omega < 1$: $\omega^n \to 0$, so $X_n(\omega) \to \omega = X(\omega)$.
- $\omega = 1$: $X_n(1) = 2$ for all n but $X(1) = 1$. No convergence.

$A = [0,1)$ has measure 1, so $X_n \xrightarrow{\text{a.s.}} X$.

Aside: the uniform distribution

Here $X(\omega) = \omega$, and for $0 \leq a < b \leq 1$,

$$\mathbb{P}(a < X < b) = \mathbb{P}\big((a,b)\big) = b - a.$$

This is the **uniform distribution** on $[0,1]$: constant density 1, so the area over (a,b) is $b-a$. (Distributions come later.)

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Why This Is Needed

Setting

$(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, and $X, Y, X_1, X_2, \ldots$ are random variables on it.

Goal

Whenever we write $\mathbb{P}$(some event described through the X_n), that event must belong to $\mathcal{F}$. We now check this for the events and variables that will appear in the theory.

Convention

In this lesson sups, infs and limits are assumed **finite** (real-valued). To allow $\pm\infty$ one works with extended-real-valued random variables; the arguments are the same.

Supremum and Infimum of a Sequence

Claim 1: $\sup_n X_n$ is a random variable

For $x \in \mathbb{R}$: $\sup_n X_n \le x$ iff **every** $X_n \le x$. So

$$\left\{\sup_n X_n \le x\right\} = \bigcap_{n=1}^{\infty} \{X_n \le x\} \in \mathcal{F},$$

a countable intersection of sets in $\mathcal{F}$.

Claim 2: $\inf_n X_n$ is a random variable

Use $\inf_n X_n = -\sup_n(-X_n)$. Each $-X_n$ is a random variable (Lesson 19), so $\sup_n(-X_n)$ is one by Claim 1, and so is its negative.

Direct version for the infimum

$\{\inf_n X_n < x\} = \bigcup_n \{X_n < x\} \in \mathcal{F}$. Note the **strict** inequality: an infimum need not be attained, so $\{\inf_n X_n \le x\}$ is not simply $\bigcup_n \{X_n \le x\}$.

lim sup and lim inf

Definitions (sequences of real numbers)

$$\limsup_n X_n = \inf_{n\geq 1} \sup_{m\geq n} X_m, \qquad \liminf_n X_n = \sup_{n\geq 1} \inf_{m\geq n} X_m.$$

Claim 3: both are random variables

For each n, $Y_n = \sup_{m\geq n} X_m$ is a random variable by Claim 1, and $\limsup_n X_n = \inf_n Y_n$ is a random variable by Claim 2. Likewise, $Z_n = \inf_{m\geq n} X_m$ is a random variable and $\liminf_n X_n = \sup_n Z_n$ is one too.

Set-level picture

Heuristically sup $\leftrightarrow \bigcap$ and inf $\leftrightarrow \bigcup$:

$$\left\{\limsup_n X_n < x\right\} \approx \bigcup_n \bigcap_{m\geq n} \{X_m < x\}.$$

Everything is built from countable unions and intersections of sets in $\mathcal{F}$.

Claim 4: Limits Are Random Variables

Claim

If $X_n(\omega)$ converges (to a real number) for **every** $\omega \in \Omega$, then $X(\omega) = \lim_n X_n(\omega)$ is a random variable.

Proof

For a sequence of real numbers, the limit exists iff $\limsup = \liminf$, and then the limit equals this common value. So

$$X = \limsup_n X_n = \liminf_n X_n,$$

and $\limsup_n X_n$ is a random variable by Claim 3. □

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Claim 5: $\{X = Y\} \in \mathcal{F}$

Proof

Earlier, $\{X > Y\} \in \mathcal{F}$, and by renaming variables $\{X < Y\} \in \mathcal{F}$. The set where X and Y differ is their union, and equality is its complement:

$$\{X = Y\} = \big(\{X > Y\} \cup \{X < Y\}\big)^c \in \mathcal{F}.$$

□

Claim 6: the convergence set is an event

Let

$$C = \big\{\omega : \lim_n X_n(\omega) \text{ exists}\big\} = \Big\{\liminf_n X_n = \limsup_n X_n\Big\}.$$

Both sides of the equality are random variables (Claim 3), so $C \in \mathcal{F}$ by Claim 5. □

Consequence

Statements such as $\mathbb{P}(\lim_n X_n \text{ exists}) = 1$ are legitimate: the set involved is measurable.

Claim 7: Gluing Two Random Variables

Claim

Let X, Y be random variables and $A \in \mathcal{F}$. Define

$$Z(\omega) = \begin{cases} X(\omega), & \omega \in A, \\ Y(\omega), & \omega \notin A. \end{cases}$$

Then Z is a random variable.

Proof

Fix $z \in \mathbb{R}$. Split Ω into A and A^c:

$$\{Z \leq z\} = \big(\{Z \leq z\} \cap A\big) \cup \big(\{Z \leq z\} \cap A^c\big) = \big(\{X \leq z\} \cap A\big) \cup \big(\{Y \leq z\} \cap A^c\big).$$

$\{X \leq z\}, \{Y \leq z\}, A, A^c \in \mathcal{F}$, and $\mathcal{F}$ is closed under finite unions and intersections, so $\{Z \leq z\} \in \mathcal{F}$. □

Special case

Take $A = C$ (the convergence set) and Y any fixed random variable: $Z = \lim X_n \,/C$, and $Z = Y \,(C)$. This is a random variable even if X_n converges only almost surely.

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Theorem: A Criterion

Notation

For $\varepsilon > 0$ let $A_n^\varepsilon = \{\omega : \ |X_n(\omega) - X(\omega)| > \varepsilon\} \in \mathcal{F}$, and

$$\limsup_n A_n^\varepsilon = \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m^\varepsilon = \{A_n^\varepsilon \text{ i.o.}\}.$$

Theorem

$$X_n \xrightarrow{\text{a.s.}} X \iff \mathbb{P}\big(A_n^\varepsilon \text{ i.o.}\big) = 0 \quad \text{for every } \varepsilon > 0.$$

Meaning

$\omega \in \{A_n^\varepsilon \text{ i.o.}\}$ iff for arbitrarily large n we have $|X_n(\omega) - X(\omega)| > \varepsilon$, so that no finite cut-off keeps the distance below ε. These are the ω's "that went bad"; the theorem says they form a null set.

Proof of the Criterion

Step 1: describe the bad set

Let $A = \{X_n \to X\}$. Then $\omega \notin A$ iff there is $\varepsilon > 0$ with $|X_n(\omega) - X(\omega)| > \varepsilon$ for infinitely many n. Taking $\varepsilon = 1/k$ suffices, so

$$A^c = \bigcup_{k=1}^{\infty} \limsup_n A_n^{1/k}.$$

Step 2: ($\Rightarrow$)

If $\mathbb{P}(A) = 1$ then $\mathbb{P}(A^c) = 0$. Each $\limsup_n A_n^{1/k} \subseteq A^c$, so it has probability 0. For arbitrary $\varepsilon > 0$ choose k with $1/k \le \varepsilon$: then $A_n^{\varepsilon} \subseteq A_n^{1/k}$ for all n, so $\limsup_n A_n^{\varepsilon} \subseteq \limsup_n A_n^{1/k}$ and has probability 0.

Step 3: ($\Leftarrow$)

If $\mathbb{P}(\limsup_n A_n^{\varepsilon}) = 0$ for every ε, then in particular for each $\varepsilon = 1/k$. By countable subadditivity, $\mathbb{P}(A^c) \le \sum_k 0 = 0$, so $\mathbb{P}(A) = 1$. $\square$

Theorem: A Sufficient Condition

Theorem

If for every $\varepsilon > 0$

$$\sum_{n=1}^{\infty} \mathbb{P}\big(|X_n - X| > \varepsilon\big) < \infty,$$

then $X_n \xrightarrow{\text{a.s.}} X$. This is **not** an "if and only if".

Proof

Recall **Borel–Cantelli I** (Lesson 11): if $E_n \in \mathcal{F}$ and $\sum_n \mathbb{P}(E_n) < \infty$, then $\mathbb{P}(\limsup_n E_n) = 0$.

Apply it with $E_n = A_n^{\varepsilon}$. The hypothesis gives $\mathbb{P}(A_n^{\varepsilon} \text{ i.o.}) = 0$ for every $\varepsilon > 0$. By the criterion, $X_n \xrightarrow{\text{a.s.}} X$. □

Why it is useful

Once the probabilities are computed, the question becomes the convergence of a **series of real numbers**, which is familiar territory.

A Version with $\varepsilon = 1/n$

Theorem

If

$$\sum_{n=1}^{\infty} \mathbb{P}\Big(|X_n - X| > \tfrac{1}{n}\Big) < \infty,$$

then $X_n \xrightarrow{\text{a.s.}} X$.

Proof

Fix $\varepsilon > 0$ and choose M with $1/M < \varepsilon$. For $n \geq M$ we have $1/n \leq 1/M < \varepsilon$, so

$$\{|X_n - X| > \varepsilon\} \subseteq \{|X_n - X| > \tfrac{1}{n}\},$$

and by monotonicity of $\mathbb{P}$ the probabilities satisfy the same inequality. Then

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n^{\varepsilon}) \leq \underbrace{\sum_{n=1}^{M-1} \mathbb{P}(A_n^{\varepsilon})}_{\leq M-1} + \sum_{n=M}^{\infty} \mathbb{P}\big(|X_n - X| > \tfrac{1}{n}\big) < \infty.$$

This holds for every $\varepsilon > 0$, so the previous theorem gives $X_n \xrightarrow{\text{a.s.}} X$. □

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Summary

Key results

Concept / Result	Content
Examples	$\omega^n \xrightarrow{\text{a.s.}} 0$ and $\omega^n + \omega \xrightarrow{\text{a.s.}} \omega$ on $[0,1]$; failure only at $\omega = 1$
sup, inf	$\{\sup X_n \le x\} = \bigcap_n \{X_n \le x\}$; $\inf = -\sup(-X_n)$
lim sup, lim inf, lim	random variables; the limit is $\limsup = \liminf$
Events	$\{X = Y\} \in \mathcal{F}$, $\{\lim X_n \text{ exists}\} \in \mathcal{F}$
Gluing	X on A, Y on A^c is a random variable
Criterion	$X_n \xrightarrow{\text{a.s.}} X \iff \mathbb{P}(\lvert X_n - X\rvert > \varepsilon \text{ i.o.}) = 0$ for all $\varepsilon > 0$
Sufficient (BC I)	$\sum_n \mathbb{P}(\lvert X_n - X\rvert > \varepsilon) < \infty \ \forall \varepsilon \Rightarrow X_n \xrightarrow{\text{a.s.}} X$
Sufficient $(1/n)$	$\sum_n \mathbb{P}(\lvert X_n - X\rvert > 1/n) < \infty \Rightarrow X_n \xrightarrow{\text{a.s.}} X$

Exercises

1. For $X_n(\omega) = \omega^n + \omega$ on $[0,1]$, check measurability by computing $\{X_n \le x\}$.
2. Show that $\{\inf_n X_n < x\} = \bigcup_n \{X_n < x\}$, and explain why $\{\inf_n X_n \le x\} \neq \bigcup_n \{X_n \le x\}$ in general.
3. Show $\{\liminf X_n < x\} \in \mathcal{F}$ directly using unions and intersections of sets $\{X_m < y\}$.
4. Let $X_n = I_{[0,1/n)}$ on $[0,1]$ with Lebesgue measure. Find the limit and show $X_n \xrightarrow{\text{a.s.}} 0$, using the criterion.
5. Show that $\sum_n \mathbb{P}(|X_n - X| > \varepsilon) < \infty$ for all ε does **not** follow from $X_n \xrightarrow{\text{a.s.}} X$ in general. (Hint: look for X_n with $\mathbb{P}(X_n = 1) = 1/n$ and independent events.)
6. Show that $\{X \le Y\} \in \mathcal{F}$ and that $\{\omega : X(\omega) \neq Y(\omega)\} \in \mathcal{F}$.
7. Prove that the $1/n$-version of the sufficient condition implies the all-ε version.

Outlook

Next lesson

Statements that are **not** "if and only if": counterexamples showing that the converse implications fail. Then **convergence in probability**, and its relationship to almost sure convergence.

Looking ahead

From now on we will not stop to verify that sets like $\{|X_n - X| > \varepsilon\}$ are measurable. You now know how to check it.